ChatGPT how AI agents can complement or help modern search or change its direction. These are the areas where tools like generative AI will reshape the search space:

- *Intuitive Search*—Search queries will become more conversational and intuitive. Generative AI models like GPT-4 have an advanced understanding of natural language, enabling them to interpret complex queries more effectively. Users will no longer need to rely on specific keywords or phrases; they can simply ask questions as they would in a conversation with another person. This will allow for more accurate and relevant search results, as AI can better comprehend the context and intent of the query. With the introduction and release of highly capable generative AI agents and models, there is a greater scope to significantly reshape full-text search capabilities provided by modern search engines like Elasticsearch. We can expect to see several key changes that will redefine the search experience for both users and developers as this technology is increasingly integrated into search platforms.
- *Personalized search*—With the incorporation of generative AI, search results can become more personalized and adaptive. Search engines will be able to learn many valuable data points from users' preferences, behavior, and search history, which in turn will help engines tailor results to meet individual users' needs. As the AI gathers more data, it will continually refine its understanding of what users are looking for, leading to an increasingly customized search experience.
- *Predictive search*—Generative AI has the potential to make search engines more proactive in anticipating user needs. Instead of merely responding to queries, AI-driven search engines may be able to predict what information users are interested in, based on their previous interactions or current context. This will allow search platforms to proactively offer relevant suggestions, increasing the value of the search experience and reducing the need for users to perform additional queries.
- *Advanced search*—Generative AI will enable search engines to provide more diverse and rich search results. By understanding the context and semantics of a query, AI-driven search engines can generate content summaries and relevant visualizations and even synthesize new information to help answer a user's question. This will lead to a more comprehensive and informative search experience that goes beyond merely linking to existing content.

In my mind, the introduction of generative AI will revolutionize full-text search capabilities, making search engines more conversational, personalized, adaptive, and proactive. This will not only enhance the user experience but also offer new opportunities for businesses and developers to create innovative search applications and services. Teams are working hard to adopt the upcoming changes in the search space with the advent of AI. So, expect a search space revolution!

This chapter laid the groundwork for using Elasticsearch by introducing its search capabilities and looking at how searching has become an integral part of numerous applications. In the next chapter, we install, configure, and run Elasticsearch and